

## CHEONAN, Korea, Republic of

WMO: 471450

Lat: 36.7625N

Lon: 127.2928E

Elev: 84

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 8.8        | 12.7 | -4.8                          | 4.3 | 18.8 | -0.8 | 5.3 | 19.5 | 15.0                  | 26.6 | 13.8 | 27.7 | 2.4                   | 270  | 0.392 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 13.7                   | 90.5            | 76.3 | 88.3 | 75.2 | 86.3 | 74.0 | 78.9                | 85.8 | 77.8 | 84.5 | 76.8 | 83.2 | 5.4                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 77.0                            | 141.2 | 82.5 | 75.9 | 136.0 | 81.6 | 74.9 | 131.1 | 80.8 | 42.5          | 86.1 | 41.4 | 84.5 | 40.4 | 83.3 | 85.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 14.3              | 12.2 | 10.7 | DB                         | 3.7 | 94.1               | 5.0 | 2.1                                                | 0.1  | 95.6       | -2.8 | 96.9       | -5.7 | 98.0       | -9.3 | 99.6 |
|                   |      |      | WB                         | 2.9 | 80.4               | 4.7 | 2.0                                                | -0.5 | 81.8       | -3.3 | 82.9       | -5.9 | 84.1       | -9.3 | 85.5 |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 54.5   | 27.0  | 33.0  | 42.5  | 53.1  | 64.6  | 72.5  | 78.0  | 78.7  | 69.4  | 57.6  | 45.1  | 31.4 |
|                                                                               | DBStd       | 18.77  | 7.57  | 7.29  | 6.72  | 6.09  | 4.73  | 3.50  | 3.47  | 4.40  | 4.60  | 6.01  | 8.02  | 7.71 |
|                                                                               | HDD50       | 2262   | 712   | 479   | 252   | 39    | 0     | 0     | 0     | 0     | 0     | 13    | 190   | 577  |
|                                                                               | HDD65       | 5084   | 1177  | 897   | 697   | 357   | 64    | 1     | 0     | 0     | 13    | 239   | 597   | 1042 |
|                                                                               | CDD50       | 3916   | 0     | 2     | 20    | 132   | 454   | 676   | 869   | 890   | 583   | 247   | 43    | 0    |
|                                                                               | CDD65       | 1262   | 0     | 0     | 0     | 0     | 53    | 227   | 404   | 425   | 145   | 8     | 0     | 0    |
|                                                                               | CDH74       | 10302  | 0     | 0     | 2     | 42    | 552   | 1577  | 3365  | 3924  | 803   | 36    | 1     | 0    |
|                                                                               | CDH80       | 3005   | 0     | 0     | 0     | 2     | 80    | 370   | 1015  | 1420  | 118   | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 4.0    | 3.9   | 4.4   | 5.2   | 5.1   | 4.4   | 3.8   | 3.7   | 3.6   | 3.4   | 3.3   | 3.9   | 3.9  |
| Precipitation                                                                 | PrecAvg     | 49.9   | 0.8   | 1.1   | 1.7   | 3.0   | 3.5   | 5.8   | 11.8  | 10.9  | 6.2   | 2.1   | 1.7   | 1.1  |
|                                                                               | PrecMax     | 73.3   | 2.3   | 3.5   | 4.4   | 7.4   | 8.1   | 14.4  | 25.4  | 35.9  | 14.9  | 7.2   | 6.1   | 2.2  |
|                                                                               | PrecMin     | 29.6   | 0.0   | 0.1   | 0.3   | 0.7   | 0.6   | 0.9   | 4.3   | 2.0   | 0.6   | 0.1   | 0.2   | 0.3  |
|                                                                               | PrecStd     | 10.8   | 0.6   | 0.9   | 1.0   | 1.5   | 1.7   | 3.7   | 4.7   | 7.2   | 4.4   | 1.8   | 1.3   | 0.6  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 49.6  | 60.2  | 70.4  | 79.1  | 85.8  | 88.1  | 91.8  | 94.6  | 86.3  | 77.8  | 70.0  | 54.9 |
|                                                                               |             | MCWB   | 41.7  | 52.9  | 55.5  | 57.2  | 62.9  | 67.8  | 75.8  | 76.5  | 73.2  | 65.6  | 59.2  | 50.1 |
|                                                                               | 2%          | DB     | 44.7  | 52.9  | 65.1  | 74.0  | 82.1  | 86.0  | 89.5  | 91.7  | 83.1  | 74.3  | 64.3  | 49.7 |
|                                                                               |             | MCWB   | 38.5  | 43.5  | 51.4  | 54.9  | 61.1  | 67.6  | 75.6  | 77.1  | 70.4  | 62.3  | 56.4  | 43.5 |
|                                                                               | 5%          | DB     | 41.6  | 48.8  | 60.1  | 70.4  | 79.6  | 84.1  | 87.5  | 89.7  | 80.9  | 71.3  | 61.2  | 46.6 |
|                                                                               |             | MCWB   | 36.4  | 40.8  | 48.4  | 52.8  | 60.9  | 67.2  | 75.4  | 76.6  | 68.5  | 60.2  | 53.8  | 41.9 |
|                                                                               | 10%         | DB     | 38.7  | 44.6  | 55.8  | 66.6  | 77.1  | 82.0  | 85.7  | 87.5  | 78.8  | 68.8  | 57.6  | 43.3 |
|                                                                               |             | MCWB   | 34.1  | 37.9  | 45.8  | 52.1  | 59.6  | 67.0  | 75.1  | 76.2  | 66.7  | 58.7  | 51.0  | 38.8 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 43.4  | 55.2  | 57.3  | 60.8  | 70.3  | 74.0  | 79.7  | 81.9  | 77.0  | 69.8  | 61.9  | 50.3 |
|                                                                               |             | MCDB   | 47.6  | 58.6  | 62.3  | 72.3  | 76.8  | 80.6  | 85.9  | 89.3  | 83.9  | 74.2  | 67.2  | 54.6 |
|                                                                               | 2%          | WB     | 40.1  | 45.3  | 54.1  | 58.2  | 65.7  | 72.6  | 78.3  | 79.6  | 73.9  | 66.4  | 58.4  | 45.6 |
|                                                                               |             | MCDB   | 43.8  | 49.7  | 61.9  | 67.8  | 74.5  | 79.5  | 85.1  | 86.3  | 79.2  | 71.3  | 62.4  | 48.6 |
|                                                                               | 5%          | WB     | 37.0  | 41.5  | 50.4  | 55.9  | 63.8  | 71.4  | 77.4  | 78.5  | 72.3  | 63.0  | 54.9  | 42.0 |
|                                                                               |             | MCDB   | 40.5  | 47.7  | 58.7  | 65.2  | 73.2  | 78.9  | 83.9  | 85.2  | 77.4  | 68.0  | 59.4  | 45.8 |
|                                                                               | 10%         | WB     | 34.5  | 38.9  | 46.7  | 54.0  | 62.2  | 69.9  | 76.6  | 77.4  | 70.3  | 60.3  | 52.3  | 39.0 |
|                                                                               |             | MCDB   | 38.6  | 43.9  | 54.0  | 63.3  | 72.2  | 77.0  | 82.7  | 84.1  | 75.1  | 65.9  | 57.1  | 43.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 16.8  | 17.2  | 20.0  | 22.3  | 21.8  | 17.8  | 12.7  | 13.7  | 17.0  | 20.7  | 17.4  | 15.7 |
|                                                                               |             | MCDBR  | 18.5  | 23.0  | 26.8  | 29.9  | 26.5  | 21.5  | 16.2  | 17.3  | 19.1  | 22.7  | 21.4  | 18.2 |
|                                                                               |             | MCWBR  | 12.6  | 14.4  | 14.8  | 13.2  | 9.8   | 7.0   | 5.7   | 6.0   | 7.6   | 10.7  | 12.0  | 13.7 |
|                                                                               | 5% WB       | MCDBR  | 15.9  | 19.5  | 22.2  | 21.4  | 19.2  | 14.6  | 11.9  | 14.0  | 13.2  | 17.1  | 16.5  | 16.8 |
|                                                                               |             | MCWBR  | 11.9  | 14.0  | 13.8  | 11.9  | 9.5   | 6.6   | 5.2   | 6.1   | 6.4   | 10.0  | 11.0  | 13.8 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.440  | 0.487 | 0.571 | 0.603 | 0.615 | 0.685 | 0.639 | 0.575 | 0.501 | 0.486 | 0.471 | 0.436 |      |
|                                                                               | taud        | 1.938  | 1.863 | 1.710 | 1.689 | 1.705 | 1.627 | 1.777 | 1.929 | 2.066 | 1.979 | 1.959 | 1.967 |      |
|                                                                               | Ebn at Noon | 222    | 228   | 221   | 224   | 224   | 208   | 217   | 228   | 239   | 226   | 210   | 213   |      |
|                                                                               | Edh at Noon | 45     | 55    | 70    | 75    | 75    | 82    | 69    | 59    | 49    | 48    | 44    | 41    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 785    | 1028  | 1352  | 1570  | 1764  | 1621  | 1317  | 1374  | 1282  | 1134  | 788   | 696   |      |
|                                                                               | RadStd      | 45     | 84    | 137   | 124   | 159   | 145   | 187   | 198   | 141   | 88    | 88    | 50    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (53 neighbors) | N/A   | N/A     | -1.46  | N/A     | N/A   | N/A   | N/A         | N/A   | +157  | +85   |

Nomenclature: See separate page